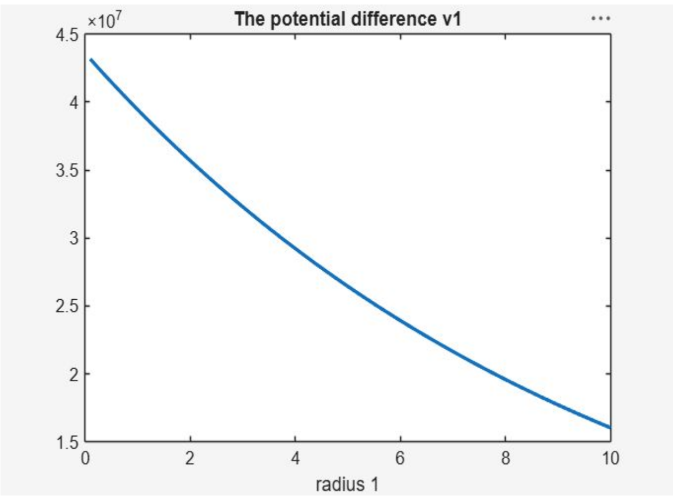
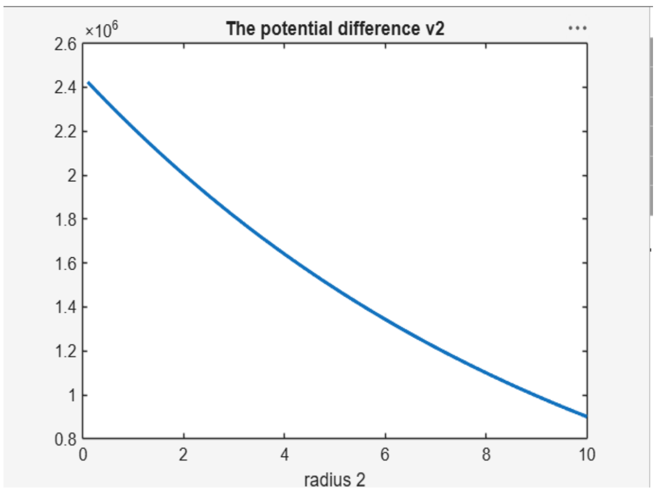
Exp. No: 5 Determination and plotting of Electric potential difference between any two points in free space medium separated by a distance R

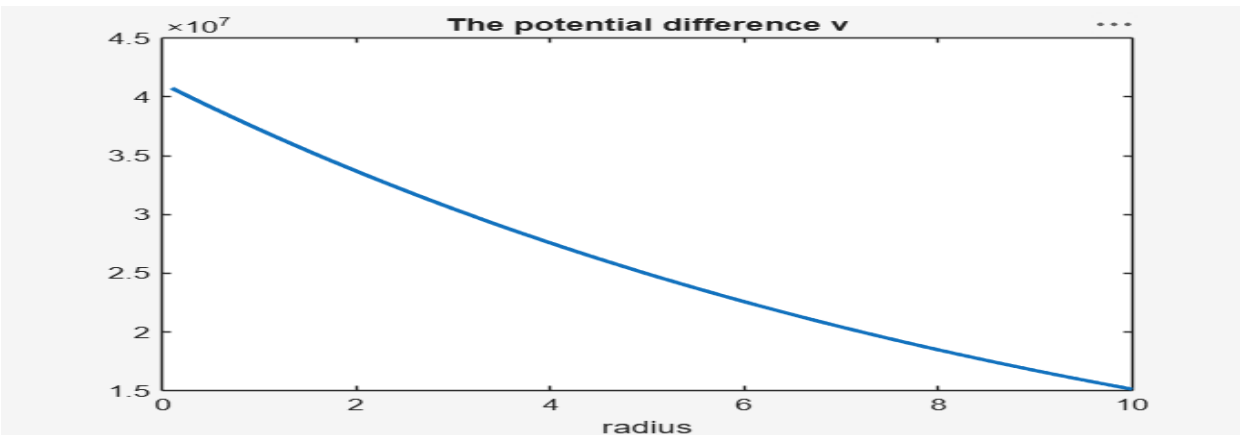
Test case (i) Determination of electric potential difference between two points for a given charge

/MATLAB Drive/Exp 5 case1.m

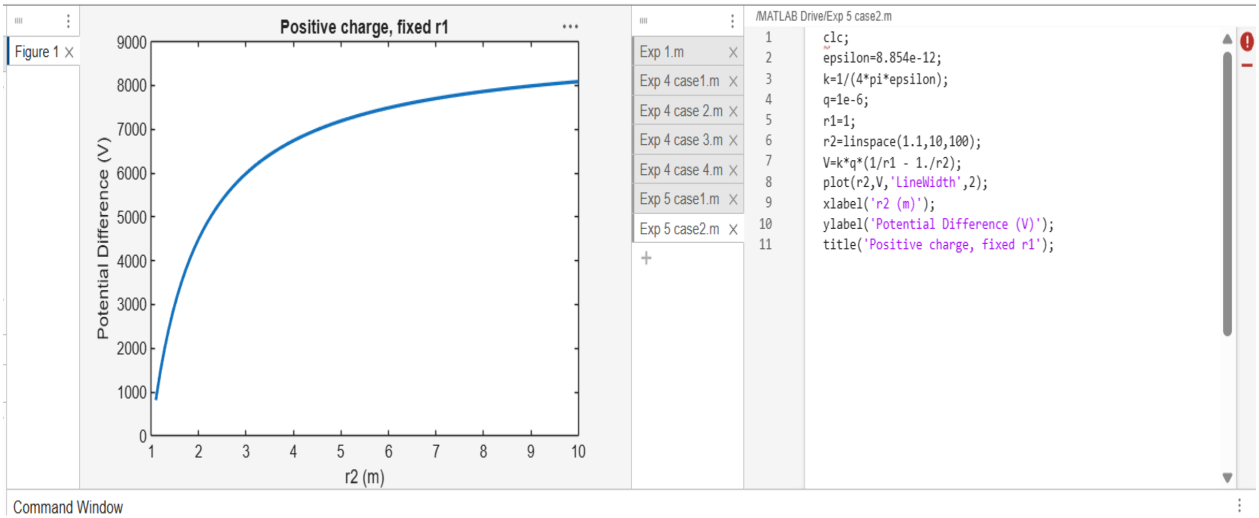
```
clc;
clear;
q=input("Enter the q value:");
epsilon=8.854*10^-12;
k=1/(4*pi*epsilon);
ax=input('Enter the value of ax=');
ay=input('Enter the value of ay=');
r1=sqrt(ax^2+ay^2);
v1=k*q/r1;
disp('The value of v1');
disp(v1);
bx=input ('Enter the value of bx=');
by=input ('Enter the value of by=');
r2=sqrt(bx^2+by^2);
v2=k*q/r2;
disp('The value of v2');
disp(v2);
v=v1-v2;
disp ('The electric potential difference is');
disp (v);
% Plot (manual style)
r = linspace(0.1,10,50);
```



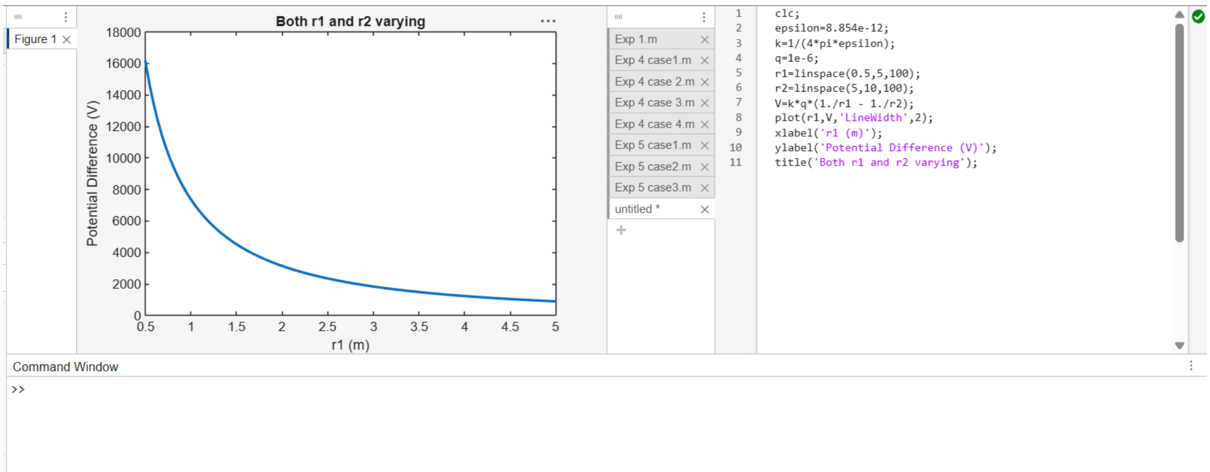




Test case (ii) A positive charge with fixed r1 = 1 m and r2 varying.



Test case (iii) A negative charge with fixed r1 and varying r2



Test case (iv) Both r1 and r2 varying with positive charge 3: